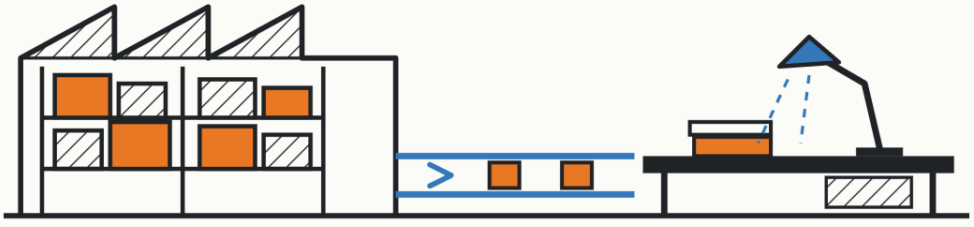
Transfers alone miss the target

Produce all, then reduce · analytical
Assumed 8 TB/s for reads + writes



Target 0.200 ms

Transfer floor ≥0.243 ms



1,856 MiB ÷ assumed 8 TB/s.
Transfers alone exceed this target.
Compute and other traffic excluded.

Scratch becomes traffic

Attention backward · one sequence, one layer · one device · FP32 partials



1,056 MiB produced

≤128 MiB retained on chip

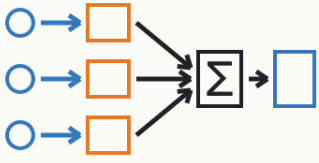
≥1,856 MiB out + back



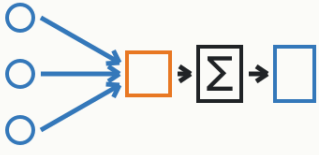
All production, then reduction.
≥928 MiB out + ≥928 MiB reread.
No compression or recomputation.

Compare the complete path

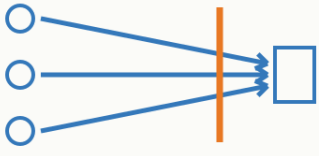
Store all partials
Simple ownership.
More scratch + traffic.



Combine smaller batches
Less live scratch.
More coordination.



Accumulate directly
Less staging.
Check synchronization.



Each design needs correctness checks.
Sum order can change rounding.